Industrial Automation & Control — Week 6 Notes

Lessons 21–25: Process Control Wrap-up, Sequence Control, PLC, RLL, Scan Cycle, Simple Programs

Exam-focused notes from scratch: compact explanations, logic flow, likely theory answers, and ladder-program reasoning. Based on your listed topics and the uploaded Week 6 assignment reference.

1. Big Picture: what these lessons are trying to teach

- **Process control / analog control** deals with continuously varying quantities such as temperature, pressure, flow, level, speed, etc.
- **Sequence control** deals with events happening in an order: start motor, wait for limit switch, energize valve, stop conveyor, raise alarm, and so on.
- **PLC** is the practical industrial controller used to implement sequence control reliably.
- **RLL (Relay Ladder Logic)** is the programming language style that looks like relay circuits and is easy for plant technicians to understand.

Memory hook
Analog/process control asks: "How much?"
Sequence control asks: "What next?"
PLC is the machine that executes the logic.
RLL is the language used to describe that logic.

2. Process-control concluding points you must know

Topic	Exam-ready meaning
RS-232	Point-to-point serial communication standard. Short distance, lower noise immunity, common in older devices and simple controller-computer links.
RS-485	Differential serial communication standard. Better noise immunity, longer distance, supports multi-drop network, very common in industrial communication.
Plant dynamics	How the process itself responds with time. A controller can never be designed properly unless the plant behaviour is understood first.
Control algorithm	The mathematical law that decides control action from error and measured variables. Example: PID equation.
Control hardware	Sensors, actuators, transmitters, I/O modules, PLC,

	controller cards, power supplies, communication interfaces.
Control software	Logic, sequencing, PID blocks, alarms, HMI screens, SCADA code, interlocks, tuning parameters.
Control performance	How well the system tracks, rejects disturbances, stays stable, avoids overshoot, and reaches the desired operating point.
Degree of freedom (DOF)	Number of independent paths by which reference, disturbance, or noise can be shaped by the controller structure.

3. Two common PID forms mentioned in process control

- **First / ideal or interacting form** controller output is written as Kc [e + (1/Ti)∫e dt + Td de/dt]. Integral and derivative are tied to the proportional gain.
- **Second / parallel or non-interacting form** controller output is Kp e + Ki∫e dt + Kd de/dt. Here Kp, Ki, and Kd look like separate gains.
- **Why two forms exist** because different manufacturers and textbooks parameterize controllers differently, but both represent the same basic P + I + D ideas.
- **Implementation point** in real hardware/software, derivative is usually filtered and often applied to measurement change rather than pure error to reduce kicks and noise amplification.

What to write in exam if asked “why implementation matters?”
The ideal PID equation is a mathematical law; the real controller must still face actuator limits, noise, sensor delay, sampling, communication delay, and mode switching.
Therefore practical PID implementation includes output limits, anti-windup, derivative filtering, and smooth auto/manual transfer.

4. Degree of freedom in control structure

- **One-degree-of-freedom control** mainly shapes tracking and disturbance rejection with one controller path. Simple and common.
- **Two-degree-of-freedom control** allows separate tuning for set-point response and disturbance rejection, so one can reduce overshoot to set-point changes without ruining disturbance response.
- **Why it matters** industrial plants often care about two different things: smooth set-point tracking and strong rejection of load disturbances. More DOF gives extra freedom in design.

If asked “how does higher degree of freedom help?”

1. Start with the problem: one controller may not give the best tracking and best disturbance rejection simultaneously.
2. Explain that extra structure adds another independent shaping path.
3. State the benefit: better trade-off between speed, overshoot, and disturbance rejection.

5. Sequence control vs analog/process control

Aspect	Analog / continuous-variable mode	Sequence / discrete-event mode
Nature of variable	Continuously varying value	Event or condition changes state
Examples	Temperature, pressure, flow, speed	Motor ON/OFF, valve open/close, start/stop sequence
Main objective	Maintain value near set-point	Execute correct order of actions
Typical model	Differential-equation / transfer-function view	Logic, state, event, sequencing view
Typical controller	PID / process controller / analog loop	PLC with ladder logic / sequence chart
Typical question	How much correction is needed?	Which output should turn on now?

Fast distinction
Continuous-variable (CV) mode → magnitude matters continuously.
Discrete-event (DE) mode → state changes matter when events occur.
Many industrial systems combine both: a plant may use PID for temperature and PLC sequence logic for startup/shutdown/interlocks.

6. Relay logic, PLC and RLL from scratch

- **Relay logic** is the older hardwired logic built using relay coils and contacts. It physically implements AND, OR, NOT, latching, interlocking, and sequencing.
- **PLC** replaced large relay panels with software-based logic. Inputs are read, program is executed, and outputs are updated cyclically.
- **RLL** copies the relay-logic style in software. Two vertical rails represent power flow; horizontal rungs represent logic conditions that energize outputs.
- **Why ladder logic became popular** because electricians and technicians can read it quickly and map software logic to physical wiring and plant devices.

Symbol / term	Meaning in logic
NO contact	Normally open contact. Conducts when the referenced input/bit is ON or true.
NC contact	Normally closed contact. Conducts when the referenced input/bit is OFF or false.
Coil	Output instruction. Energizes a relay/output/bit when the rung becomes true.
Auxiliary contact	Extra contact associated with a relay or output bit, used for self-hold, interlock, or status feedback.
Emergency stop	Hard safety stop input, usually wired fail-safe and treated carefully; often represented as NC logic so wire break also causes stop.
Interlock	Logic that prevents dangerous or conflicting outputs from turning on together.

7. Scan cycle — the heart of PLC operation

- **Step 1: Read inputs** PLC samples all physical input states and stores them in an input image table.
- **Step 2: Execute program** Rungs are solved from top to bottom and left to right using the stored image, not continuously re-reading the hardware in the middle of the rung.
- **Step 3: Update outputs** PLC writes the computed output image to output modules.
- **Step 4: Housekeeping / communication** diagnostics, communication, memory management, then next scan begins.
- **Key consequence** if an input changes after the input-scan stage, the program generally reacts only in the next scan unless special immediate instructions are used.

One-line definition
A PLC does not think continuously; it thinks in repeated scan cycles.

How to reason about scan-cycle questions

4. First ask: what input states were captured in the input image at the beginning of the scan?
5. Then solve each rung in order using those states.
6. Finally decide which outputs will be written at the end of that scan.

8. Logic and program flow in simple RLL programs

- **Series contacts implement AND logic** because all contacts in the path must conduct for power flow to reach the coil.
- **Parallel branches implement OR logic** because any one conducting branch can complete the path.
- **NC contact acts like logical NOT of the bit state** because it conducts when the input/bit is false.
- **A seal-in / latch circuit** uses an auxiliary contact of the output in parallel with the start pushbutton so the output remains energized after the start button is released.
- **Interlock rung** ensures that if one machine direction/output is ON, the opposite output cannot turn on.
- **Program flow instructions** such as jump to subroutine (JSR), subroutine (SBR), and return (RET) change the order in which parts of the program are executed.

Pattern	Meaning / exam explanation
Start PB in parallel with output auxiliary contact, stop PB in series	Standard motor start-stop holding circuit.
Two opposite outputs each using NC contact of the other in series	Electrical interlock. Prevents simultaneous actuation.
NC emergency stop ahead of control rung	Fail-safe stop path. If E-stop is pressed or wiring fails open, rung drops out.
JSR -> SBR -> RET	Main program can call a subroutine, execute the logic inside it, and return to the main program afterwards.

9. How to solve ladder-logic questions in exam

Universal method

7. Identify every contact as NO or NC.
8. Replace NO contact by the variable itself and NC contact by the complement of the variable.
9. Read series path as AND.
10. Read parallel path as OR.
11. Simplify branch by branch until you reach the output coil expression.
12. For word problems, translate physical meaning carefully: “sensor satisfactory” means contact true; “switch open” or “not pressed” may mean false depending on the symbol used.
13. For scan-cycle problems, solve in rung order using stored input image.

Typical logic translations
Series path: A in series with B → A.B
Parallel path: A in parallel with B → A + B
NC contact: NC A → $\bar{A}$ in Boolean form
Seal-in: Start + Output_aux, then series with Stop NC
Interlock: Forward . $\overline{\text{Reverse}}$ or Reverse . $\overline{\text{Forward}}$

10. Exam-ready answers from scratch

Q. What is sequence control?

Answer: Sequence control is a mode of control in which actions are carried out in a prescribed order depending on events, conditions, and logic states. The controlled variables are usually discrete states such as ON/OFF, OPEN/CLOSE, START/STOP rather than continuously varying magnitudes.

- It is used in conveyors, filling machines, motor start sequences, elevator logic, boiler startup, batch operations, and safety interlock systems.
- Its main concern is correct order, safe transition, and proper interlocking between actions.

Q. What is a PLC and why is it preferred over hardwired relays?

Answer: A PLC is a Programmable Logic Controller, a rugged industrial computer used for control and sequencing.

- It is preferred because logic changes can be made in software instead of rewiring hardware.
- It reduces panel complexity, improves reliability, makes troubleshooting easier, and allows timers, counters, communication, diagnostics, and modular I/O.

Q. Explain the PLC scan cycle.

Answer: In every scan, the PLC first reads inputs and stores them in memory, then executes the user program rung by rung, then updates outputs, and finally performs housekeeping and communication tasks.

- Hence outputs are based on the input image captured at the beginning of the scan, not on instant mid-scan physical changes.
- This cyclic operation is called the scan cycle.

Q. Differentiate sequence control and analog control.

Answer: Analog control deals with continuously varying variables and typically uses process-control algorithms like PID to maintain a set-point.

- Sequence control deals with the order of discrete actions and typically uses logical conditions, events, states, and interlocks.
- In modern plants both are often combined: PLC logic handles sequence and interlocks, while PID handles continuous-variable loops.

Q. What is an interlock? Why is it important?

Answer: An interlock is a logic or hardware arrangement that prevents an unsafe, conflicting, or undesirable action from occurring.

- For example, forward and reverse contactors of a motor should not energize together. An interlock ensures that when one is ON, the other is blocked.
- It improves safety, protects equipment, and prevents process mistakes.

Q. Why is an emergency stop often treated as a normally closed contact?

Answer: A normally closed emergency-stop path is fail-safe. During healthy operation the circuit remains closed; pressing the E-stop opens the path and removes power/logic.

- If the wire breaks or supply fails, the path also opens and the machine stops, which is safer than a normally open arrangement.

Q. What is the role of RS-232 and RS-485 in industrial control?

Answer: They are serial communication standards used to connect controllers, PLCs, instruments, HMIs, and computers.

- RS-232 is simpler and mainly point-to-point for short distance.
- RS-485 is better for industrial use because differential signalling gives higher noise immunity, longer distance, and multi-device networking capability.

11. Flow / logic behind answers — how the examiner expects you to think

For a word-based ladder question

14. Identify the physical condition that must definitely be true for the output to energize.
15. Map that condition to NO or NC contact behaviour.
16. Check whether conditions are simultaneous (series -> AND) or alternative (parallel -> OR).
17. Check whether any stop/interlock condition is inserted in series as an NC block.
18. Only then write the Boolean expression or select the diagram.

For JSR / SBR / RET questions

19. Remember that JSR is a call from main program to subroutine.
20. The logic inside the subroutine runs until RET is reached.
21. Then execution returns to the next instruction after the JSR in the main program.
22. So always ask: which outputs belong to the main program and which belong to the subroutine?

For sequential-function-chart questions

23. Read state by state, not all at once.
24. A transition becomes active only when its condition is satisfied and the preceding state is already active.
25. An output tied to a state exists only when that state is active.
26. Therefore if a later state depends on X2 and X3, the earlier state/output conditions must already have been realized first.

12. Very likely exam asks

- Define sequence control and compare it with analog/process control.
- Explain PLC, relay logic, and RLL.
- Explain the PLC scan cycle with a neat block flow.
- Distinguish NO and NC contacts in ladder logic.
- Explain auxiliary contacts, seal-in circuit, and interlocking.
- Explain why emergency stop is treated in fail-safe manner.
- Derive Boolean expression from a ladder rung or draw ladder from logic expression.
- Explain JSR, SBR, and RET in program flow.
- State applications of PLC-based sequence control.
- Short note on RS-232 vs RS-485.

1-page ultra-short revision
Sequence control = ordered actions based on events.
PLC = rugged industrial controller.
RLL = relay-style programming language.
Series = AND, parallel = OR, NC = NOT.
Scan cycle = read inputs -> execute logic -> update outputs -> housekeeping.
Auxiliary contact = used for self-hold and interlocks.
E-stop is fail-safe, often represented through NC logic path.
RS-232 = short, point-to-point; RS-485 = long, noise-resistant, multi-drop.
Analog control handles continuous variables; sequence control handles event order.
Solve ladder questions by converting rung paths into Boolean logic carefully.

Reference used: uploaded Week 6 assignment/lesson PDF for topic and question-pattern alignment.